Data Science

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Unit 4

Forecasting and Time series Analysis

Introduction

Time series analysis and forecasting are crucial for predicting future trends, behaviors, and behaviors based on historical data

It aids in planning, budgeting, and strategizing across various domains such as finance, economics, healthcare, climate science, and resource management, driving efficiency and competitiveness.

It helps businesses

- make informed decisions,
- optimize resources,
- mitigate risks by anticipating market demand,
- sales fluctuations,
- stock prices,

What is Time Series?

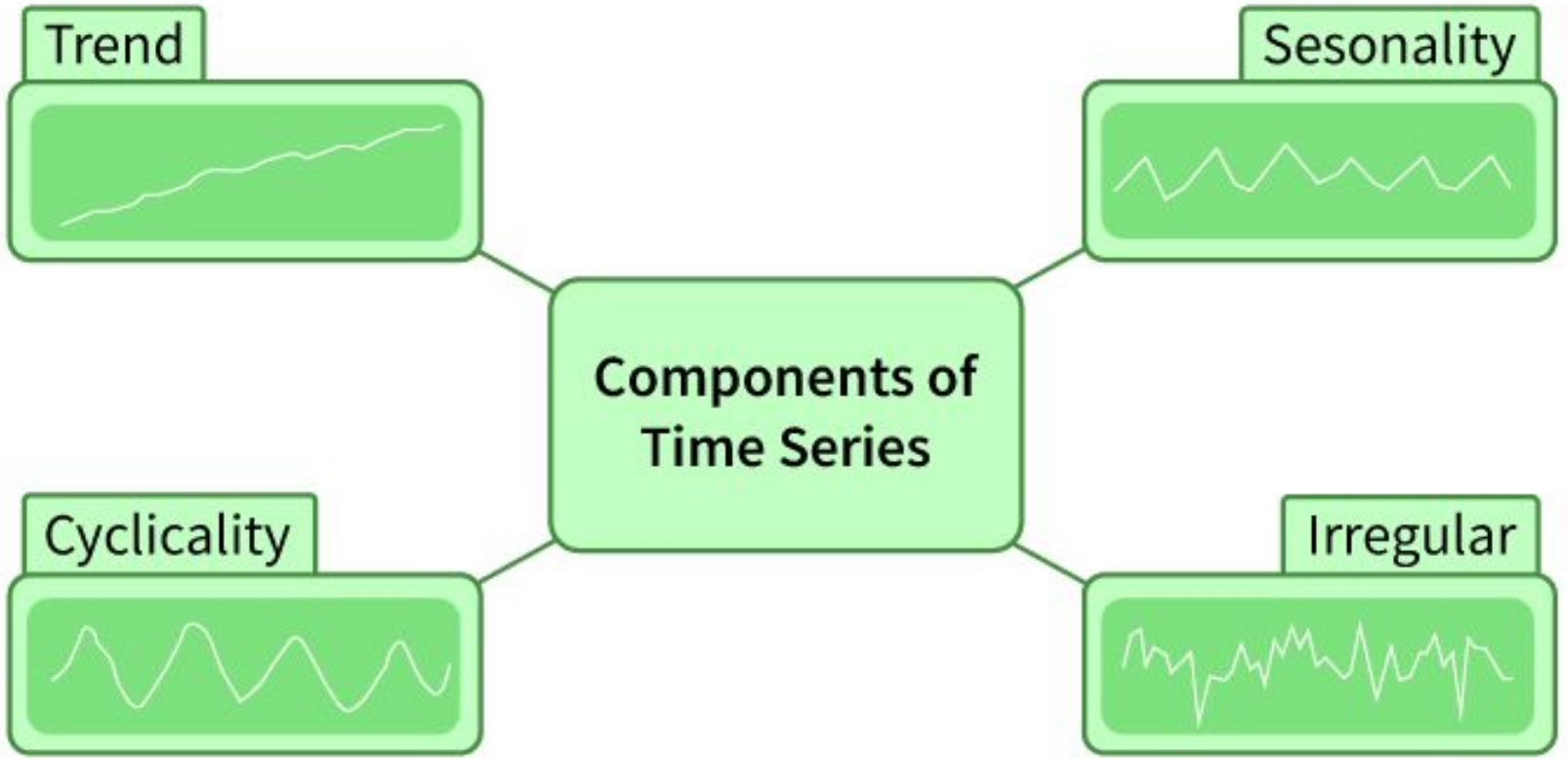
A time series is a sequence of data points collected, recorded, or measured at successive, evenly-spaced time intervals.

Each data point represents observations or measurements taken over time, such as stock prices, temperature readings, or sales figures.

Time series data is commonly represented graphically with time on the horizontal axis and the variable of interest on the vertical axis

Facilitates to identify trends, patterns, and fluctuations in the variable over time, aiding in the analysis

Components of Time Series



Trend

It is a pattern in data that shows the movement of a series to relatively higher or lower values over a long period of time

A trend is observed when there is an increasing or decreasing slope in the time series

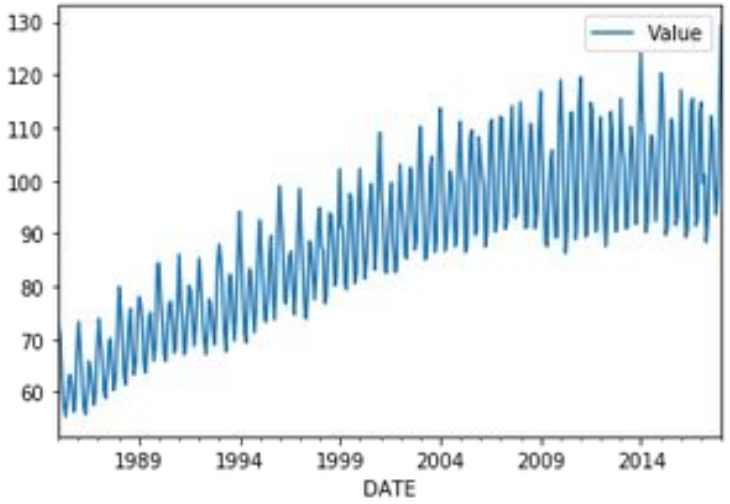
Trend usually happens for some time and then disappears, it does not repeat

Example: A new song goes trending for a while, and then disappears

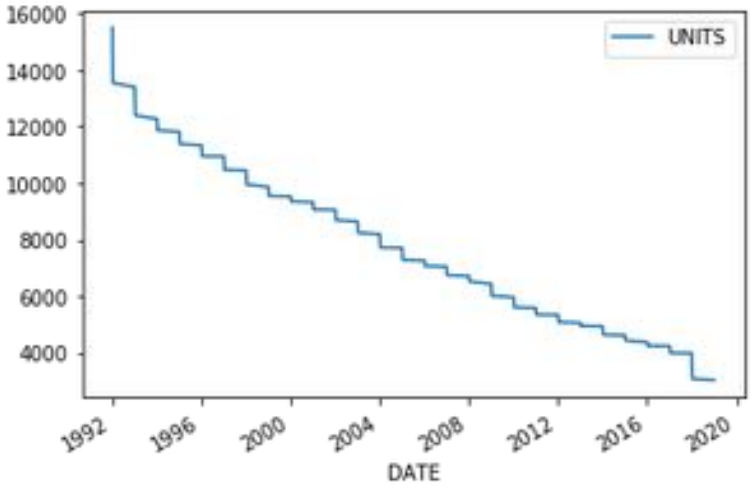
Trend Contd..

A trend could be :

Uptrend: Time Series Analysis shows a general pattern that is upward then it is Uptrend.
Downtrend: Time Series Analysis shows a pattern that is downward then it is Downtrend.
Horizontal or Stationary trend: If no pattern observed then it is called a Horizontal or stationary trend.



Uptrend



Downtrend



Horizontal trend

Seasonality

Seasonality refers to the recurring and predictable patterns that occur at regular intervals within a time series.

These patterns often follow a cyclic or periodic nature and can be influenced by various factors like weather, holidays, or business cycles.

Seasonality manifests as periodic fluctuations that repeat over fixed time intervals like days, months, or years.

Why to detect Seasonality?

Pattern **Recognition**: detects recurring patterns within a time series

Forecasting: By detecting seasonality, analysts can account for these patterns when building predictive models which leads to more robust and reliable forecasts.

Anomaly Detection: identify anomalies or irregularities in the data. Sudden deviations from the expected seasonal pattern may signal important events or changes that warrant further investigation.

Optimized Decision-Making: enables organizations to optimize resource allocation, inventory management and marketing strategies based on anticipated temporal fluctuations in demand or other relevant metrics.

Cyclic variations

Cyclical variations are longer-term fluctuations in the time series that do not have a fixed period like seasonality

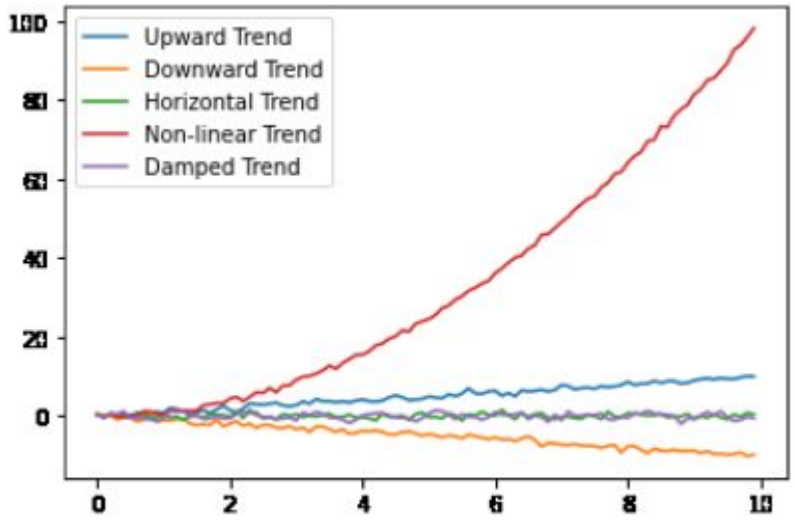
These fluctuations represent economic or business cycles, which can extend over multiple years and are often associated with expansions and contractions in economic activity.

Irregularity (or Noise)

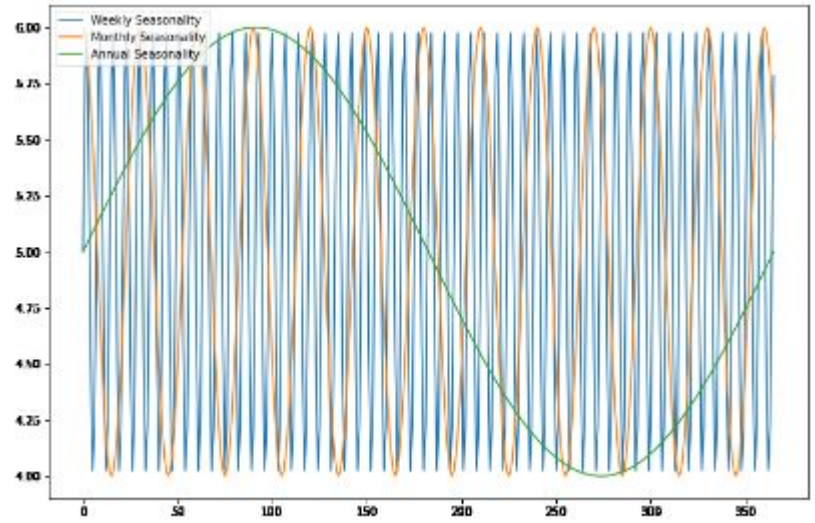
Irregularity, also known as noise or randomness, refers to the unpredictable or random fluctuations in the data that cannot be attributed to the trend, seasonality, or cyclical variations.

These fluctuations may result from random events, measurement errors, or other unforeseen factors

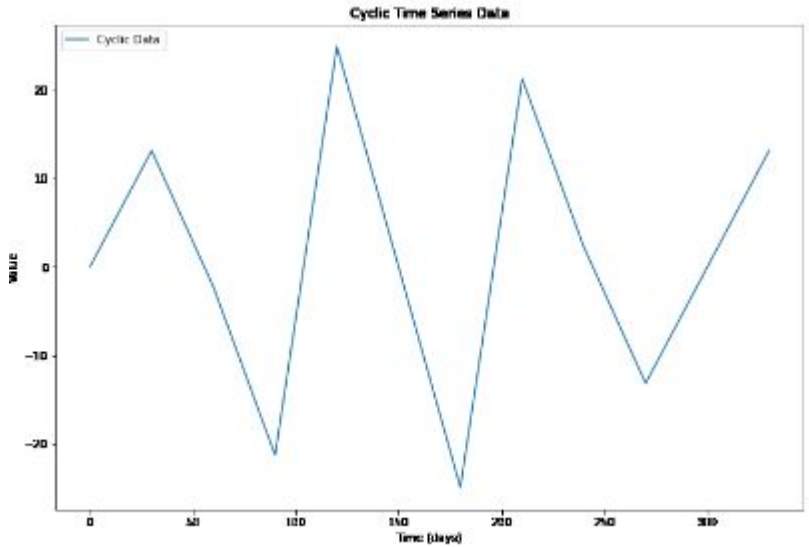
Irregularity makes it challenging to identify and model the underlying patterns in the time series data.



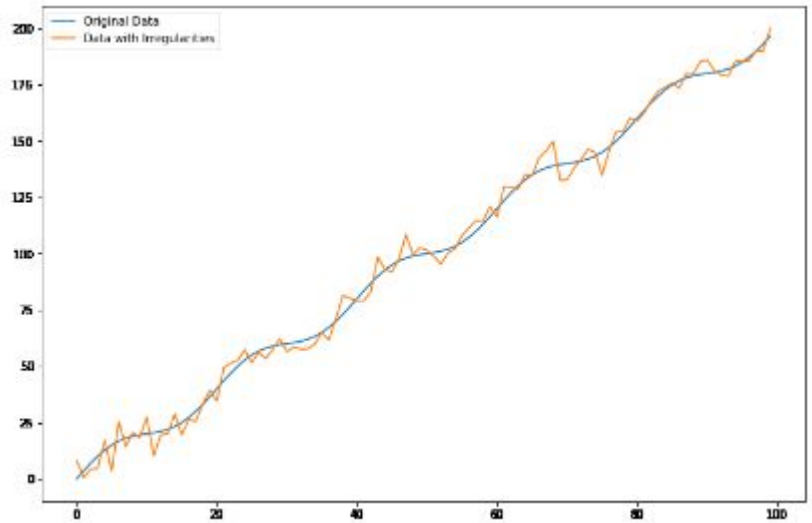
Various Trends in Time Series Data



Seasonality in Time Series Data



Cyclicity in Time Series Data



Irregularities in Time Series Data

Types of Data

Time Series Data: Comprises observations collected at different time intervals. It's geared towards analyzing trends, cycles, and other temporal patterns.

Cross-Sectional Data: Involves data points collected at a single moment in time. Useful for understanding relationships or comparisons between different entities or categories at that specific point.

Pooled Data: A combination of Time Series and Cross-Sectional data. This hybrid enriches the dataset, allowing for more nuanced and comprehensive analyses.

Some Time Series Terms

Dependence: The relationship between two observations of the same variable at different periods is crucial for understanding temporal associations.

Stationarity: A property where the statistical characteristics like mean and variance are constant over time; often a prerequisite for various statistical models.

Differencing: A transformation technique to turn stationary into non-stationary time series data by subtracting consecutive or lagged values.

Specification: The process of choosing an appropriate analytical model for time series analysis could involve selection criteria, such as the type of curve or the degree of differencing.

Some Time Series Terms

Exponential Smoothing: A forecasting method that uses a weighted average of past observations, prioritizing more recent data points for making short-term predictions.

Curve Fitting: The use of mathematical functions to best fit a set of data points, often employed for non-linear relationships in the data.

ARIMA (Auto Regressive Integrated Moving Average): A widely-used statistical model for analyzing and forecasting time series data, encompassing aspects like auto-regression, integration (differencing), and moving average.

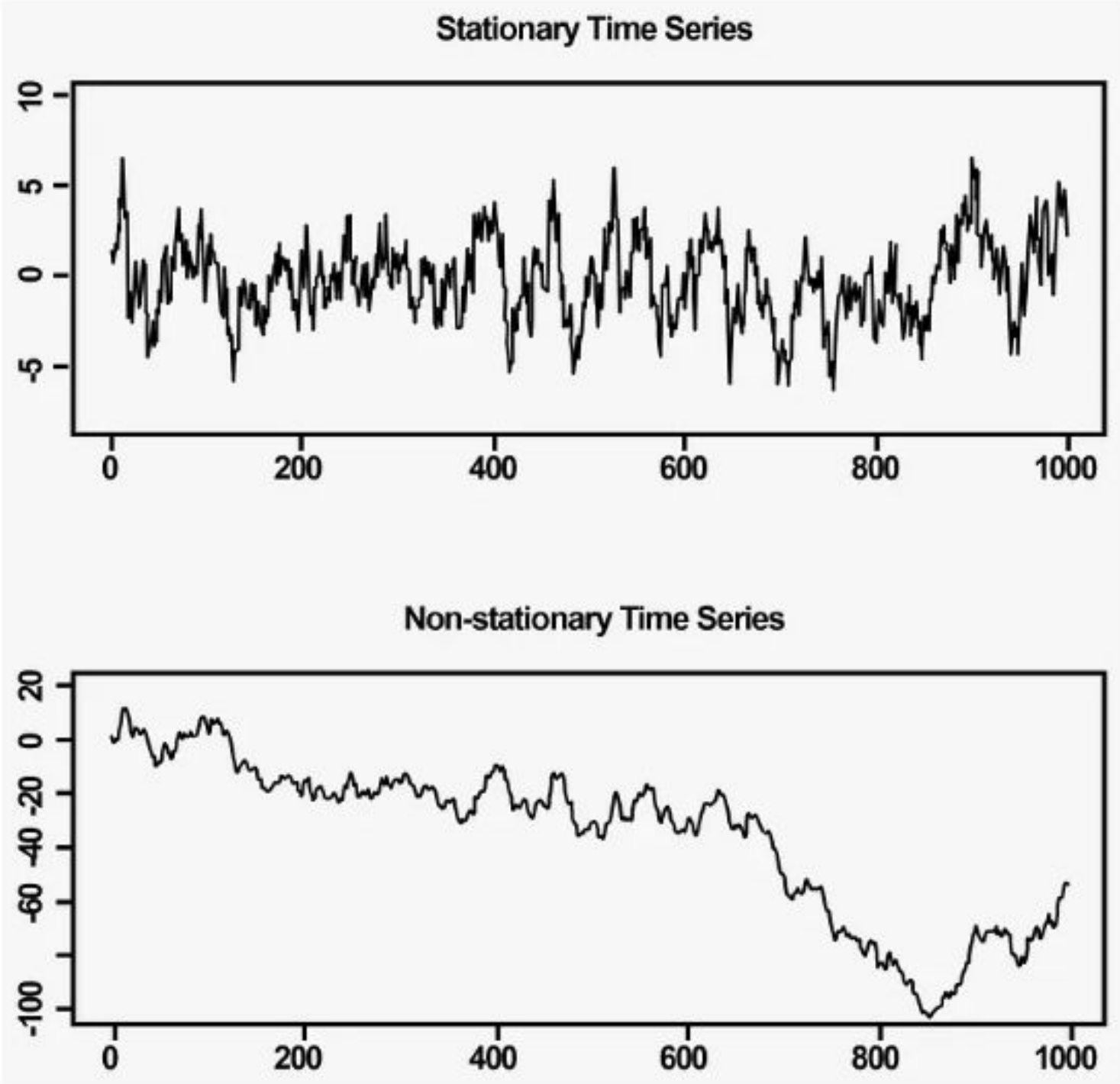
Stationarity and nonstationarity

A time series is said to be stationary when its statistical properties are constant and there's no seasonality in the time series

Time-series data is said to be stationary if it doesn't change with time or if they don't have a temporal structure.

In time series forecasting, we cannot derive valuable insights from data if it is stationary.

In a non-stationary time series, the statistical properties change over time, and there is a trend and seasonality component.


Stationary Time Series
10
5
0
-5
0
200
400
600
800
1000
Non-stationary Time Series
20
0
-20
-40
-60
-100
0
200
400
600
800
1000

For a time series to be stationary,

- The mean of the time series is constant.
- The standard deviation of the time series is constant.
- There’s no trend or seasonality in the time series.

Types of Stationarity

Trend Stationary – A time series that does not show a trend.

Seasonal Stationary – A time series that does not show seasonal changes.

Strictly Stationary – The joint distribution of observations is invariant to time shift.

Moving Averages

Average price of last N number of periods

Smoothens the short term price fluctuations of the stock

Gives a big picture of a trend

It is a calculation to analyze data points by creating a series of averages of different selections of the full data set

Autoregressive models

Captures the relationship between an observation and several lagged observations (previous time steps).

Applications in various fields, including finance, economics, climate science, etc

The core idea is that the current value of a time series can be expressed as a linear combination of its past values, with some random noise.

Autoregressive models

Mathematically, an autoregressive model of order p, denoted as AR(p), can be expressed as:

$$X_t = c + \phi_1 X_{t-1} + \phi_2 X_{t-2} + \ldots + \phi_p X_{t-p} + \varepsilon_t$$

Where, X_t is the value at time t.

c is a constant

$\phi_1, \phi_2, \ldots, \phi_p$ are the model parameters

$X_{t-1}, X_{t-2}, \ldots, X_{t-p}$ are the lagged values.

ε_t represents white noise (random error) at time t.

Autoregressive models

ARMA models are sometimes called atheoretic models- they are not derived from any economic theory

$$gdp_t = a + bgdp_{t-1} + u_t$$

$$gdp_t = a + b(Lgdp) + u_t \text{ or } (1 - bL)\, gdp = a_0 + u_t$$

$$AR(2): gdp_t = a + b_1 gdp_{t-1} + b_2 gdp_{t-2} + u_t$$

$$AR(3): gdp_t = a + b_1 gdp_{t-1} + b_2 gdp_{t-2} + b_3 gdp_{t-3} + u_t$$

$$AR(p): gdp_t = a + \sum_{i=1}^{p} b_i gdp_{t-i} + u_t$$

$$MA(2): gdp_t = \gamma + d_0 u_t + d_1 u_{t-1} + d_2 u_{t-2}$$

$$MA(q): gdp_t = \gamma + d_0 u_t + \sum_{j=1}^{q} d_j u_{t-j}$$

ARMA(*1, 1*): $gdp_t = a + b_1 gdp_{t-1} + d_0 u_t + d_1 u_{t-1}$

(model contains *1* lag each of the dependent variable and the error term)

ARMA(*p, q*): $gdp_t = a + \sum_{i=1}^{p} b_i gdp_{t-i} + d_0 u_t + \sum_{j=1}^{q} d_j u_{t-j}$

(model contains ***p*** lags of the dependent variable and ***q*** lags of the error term)

Autocorrelation (ACF)

It refers to the correlation between a time series and a lagged version of itself.

In the context of autoregressive models, autocorrelation measures how closely the current value of a time series is related to its past values, specifically those at different time lags.

Autocorrelation (ACF)

Breakdown:

Autocorrelation involves calculating the correlation between a time series and a lagged version of itself. The "lag" represents the number of time units by which the series is shifted. For example, a lag of 1 corresponds to comparing the series with its previous time step, while a lag of 2 compares it with the time step before that, and so on.

Lag values help you calculate autocorrelation, which measures how each observation in a time series is related to previous observations.

Autocorrelation (ACF)

Breakdown:

The autocorrelation at a particular lag provides insights into the temporal dependence of the data.

If the autocorrelation is high at a certain lag, it indicates a strong relationship between the current value and the value at that lag. Conversely, if the autocorrelation is low or close to zero, it suggests a weak or no relationship.

Autocorrelation (ACF)

Breakdown:

To visualize autocorrelation, a common approach is to create an ACF plot. This plot displays the autocorrelation coefficients at different lags.

The horizontal axis represents the lag, and the vertical axis represents the autocorrelation values

Autocorrelation (ACF)

Breakdown:

In an Autoregressive model of order p, the current value of the time series is expressed as a linear combination of its past p values, with coefficients determined through methods like least squares or maximum likelihood estimation

Autocorrelation (ACF)

Breakdown:

Autocorrelation can also be used to assess whether a time series is stationary.
In a stationary time series, autocorrelation should gradually decrease as the lag increases.
Deviations from this behavior might indicate non-stationarity.

Types of Autocorrelation models

AR(1) Model:

In the AR(1) model, the current value depends only on the previous value.

It is expressed as: $X_t = c + \phi_1 X_{t-1} + \varepsilon_t$

AR(p) Model:

The general autoregressive model of order p includes p lagged values.

It is expressed as shown in the introduction.

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt


# Set a random seed for reproducibility
np.random.seed(0)

# Load your temperature dataset with columns
&amp;quot;Date&amp;quot; and &amp;quot;Temperature&amp;quot;
data = pd.read_excel('Data.xlsx')

# Make sure your &amp;quot;Date&amp;quot; column is in
datetime format
data['Date'] = pd.to_datetime(data['Date'])

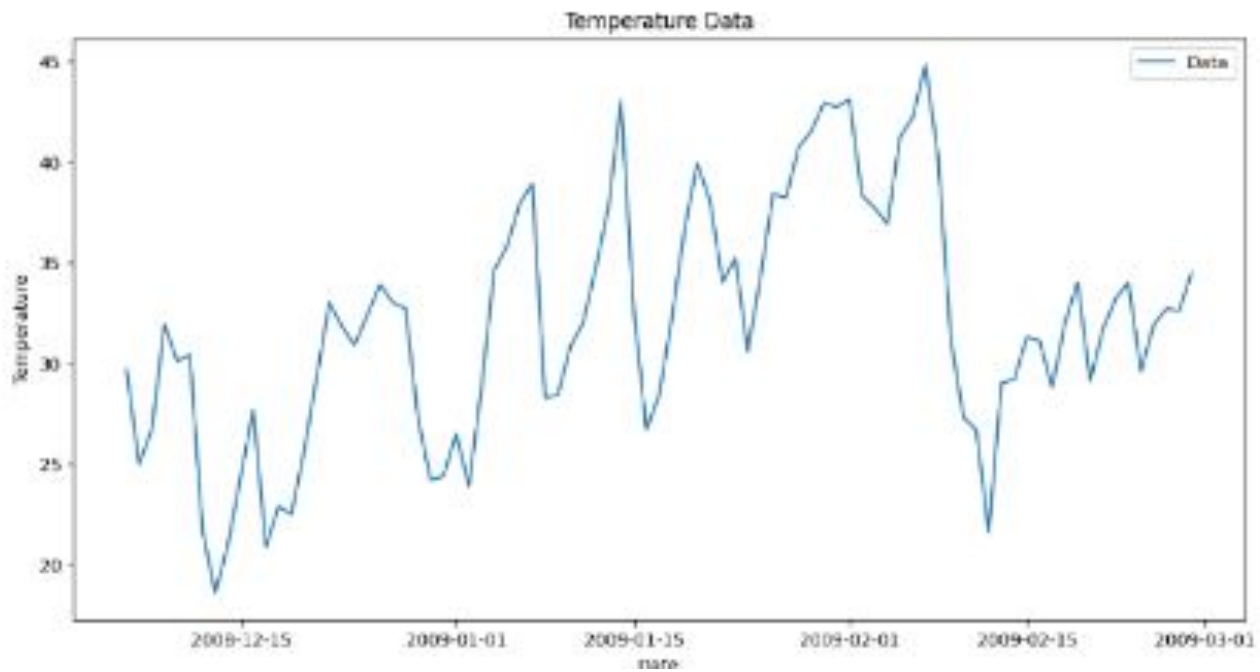
# Sorting the data by date (if not sorted)
data = data.sort_values(by='Date')

# Resetting the index
data.set_index('Date', inplace=True)

data.dropna(inplace=True)
```

```
# Visualize the data
plt.figure(figsize=(12, 6))
plt.plot( data['Temperature '], label='Data')
plt.xlabel('Date')
plt.ylabel('Temperature')
plt.legend()
plt.title('Temperature Data')
plt.show()
```

Output:



```
# Adding lag features to the DataFram
for i in range(1, 6):  # Creating lag features up
to 5 days
    data[f'Lag_{i}'] = data['Temperature
'].shift(i)

# Drop rows with NaN values resulting from
creating lag features
data.dropna(inplace=True)

# Split the data into training and testing sets
train_size = int(0.8 * len(data))
train_data = data[:train_size]
test_data = data[train_size:]

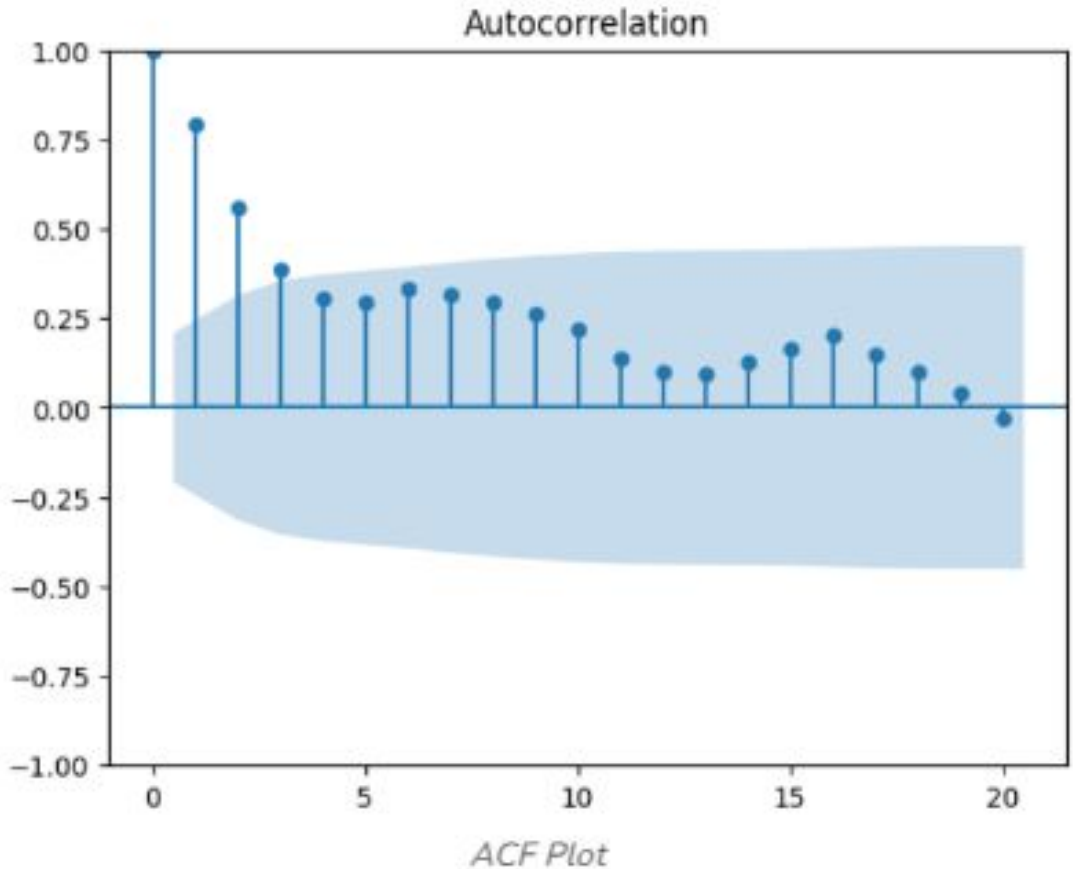
# Define the input features (lag features) and
target variable

y_train = train_data['Temperature ']

y_test = test_data['Temperature ']
```

```
from statsmodels.graphics.tsaplots import t
plot_acf
series = data['Temperature ']
plot_acf(series)
plt.show()
```

Output:



ACF Plot

data['Temperature '].corr(data['Temperature '].shift(1))

```
from statsmodels.tsa.ar_model import o
from statsmodels.graphics.tsaplots import
plot_acf
from statsmodels.tsa.api import AutoReg
from sklearn.metrics import mean_absolute_error,
mean_squared_error

# Create and train the autoregressive model
lag_order = 1 # Adjust this based on the ACF plot
ar_model = AutoReg(y_train, lags=lag_order)
ar_results = ar_model.fit()
```

```
# Make predictions on the test set
y_pred =
ar_results.predict(start=len(train_data),
end=len(train_data) + len(test_data) - 1,
dynamic=False)
#print(y_pred)

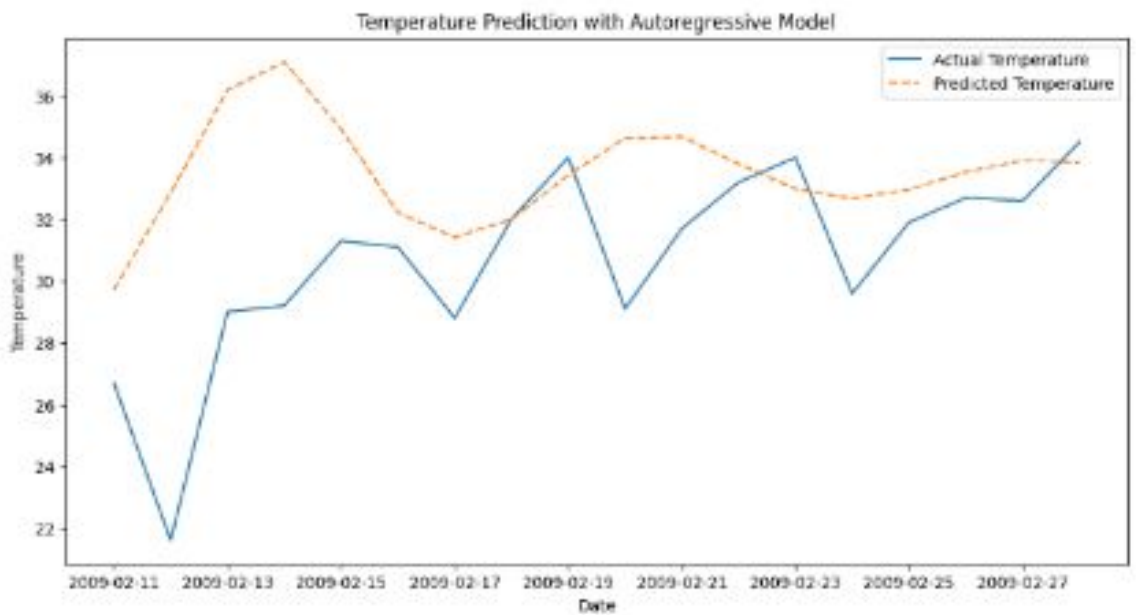
# Calculate MAE and RMSE
mae = mean_absolute_error(y_test, y_pred)
rmse = np.sqrt(mean_squared_error(y_test,
y_pred))
print(f'Mean Absolute Error: {mae:.2f}')
print(f'Root Mean Squared Error: {rmse:.2f}')
```

Output:

```
Mean Absolute Error: 1.59
Root Mean Squared Error: 2.30
```

```
# Visualize the results
plt.figure(figsize=(12, 6))
plt.plot(test_data[&amp;quot;Date&amp;quot;] ,y_test,
label='Actual Temperature')
plt.plot( test_data[&amp;quot;Date&amp;quot;],y_pred,
label='Predicted Temperature', linestyle='--')
plt.xlabel('Date')
plt.ylabel('Temperature')
plt.legend()
plt.title('Temperature Prediction with
Autoregressive Model')
plt.show()
```

Output:



ARIMA Models:

Having understood how to specify ARMA models, the concept of ARIMA will not be difficult to assimilate.

1. Distinction between ARMA and ARIMA is the ***integration*** component which brings us back to the subject of stationarity.
2. In reality, most economic variables are non-stationary hence they have to go through a transformation process called differencing before they become stationary.
3. The transforming process is also called **integration**.
4. So ARIMA informs the researcher or reader that the series in question has gone through an integration process before being used for any analysis.
5. Hence, the moment a nonstationary variable is differenced before becoming stationary, such is known as an *integrated* variable.

ARIMA Model Specification:

ARIMA *(p, d, q)* tells us the number of lags of the dependent variable (*p*), how many times the variable is differenced to become stationary (*d*) and the number of lags of the error term (*q*). Such that:

1. ARIMA*(1, 1, 2)* indicates the model has: one lag of the dependent variable (*1*), the variable being used is of first-difference stationary (*1*) and two lags of the error term (*2*)

2. ARIMA*(1, 0, 1)* indicates the model has: one lag of the dependent variable (*1*), the variable being used is of level stationary (*0*) and one lag of the error term (*1*).

3. ARIMA*(1, 0, 1)* = ARMA *(1, 1)* if the series is stationary in level.

Interpreting ACF Plots

When you look at an ACF plot, you'll see bars at each lag. The height of the bar represents the correlation coefficient at that lag.

Significant Lag: If a bar extends beyond the significance bounds, it indicates significant autocorrelation at that lag.
Gradual Decline: A gradual decline in bar heights suggests a long-term dependency in the data.
Seasonal Patterns: Regular spikes at certain lags suggest seasonality in the data.

Thank you